

VARIANTS OF STANLEY SEQUENCES

PICO GILMAN AND MARK SAENGRUNGKONGKA

ABSTRACT. Given a set of integers with no arithmetic progression of length 3, we construct a Stanley sequence by adding integers greedily so that no arithmetic progression of length 3 is formed. We discuss two different results about variants of this sequence. First, we make the sequence that avoids solutions to $x + y = z$ (instead of avoiding arithmetic progression of length 3; which amounts to avoiding solutions to $x + y = 2z$). We prove that if we start with a 2-element set, the resulting sequence $a_1, a_2, \dots$ has $a_k - a_{k-1}$ eventually periodic. Second, we disprove a conjecture of Kiss, Sándor, and Yang for all $k \geq 6$. This conjecture, if true, would have given an upper bound on greedy sequences avoiding arithmetic progressions of length k .

1. INTRODUCTION

A subset of nonnegative integers is **k -AP free** if it does not contain a nonconstant arithmetic progression of length k . Determining the size $r_k(n)$ of the largest k -AP free subset of $\{1, 2, \dots, n\}$ has been of keen interest in additive combinatorics. For example, in the case of $k = 3$, the best known lower bound (i.e., construction) is due to Behrend [Beh46], which shows that $r_3(n) \geq n \exp(-c\sqrt{\log n})$ for some constant $c > 0$. The best known upper bound is due to Kelley and Meka [KM23], which shows that $r_3(n) \leq n \exp(-c(\log n)^\beta)$ for some constants $c, \beta > 0$.

In 1978, Odlyzko and Stanley proposes a construction a 3-AP-free set via a greedy algorithm.

Definition 1. Given a 3-AP free subset $A_0 = \{a_1 < a_2 < \dots < a_t\} \subseteq \mathbb{N}_0$, the **Stanley sequence** $S_3(A_0)$ is a set $\{a_1 < a_2 < \dots\}$ such that for all $i \geq t + 1$, a_i is the smallest nonnegative integer a such that $a > a_{i-1}$ and $\{a_1, \dots, a_{i-1}, a\}$ is 3-AP free.

Example 2. We have

$$S_3(\emptyset) = S_3(\{0\}) = \{0, 1, 3, 4, 9, 10, 12, 13, 27, 28, 30, 31, 36, 37, 39, 40, 81, \dots\}.$$

Written in base 3, we have

$$S_3(\emptyset) = \{0_3, 1_3, 10_3, 11_3, 100_3, 101_3, 110_3, 111_3, 1000_3, 1001_3, 1010_3, 1011_3, 1100_3, \dots\}.$$

As the example suggests, one can easily classify the sequence: a straightforward induction argument shows that $S_3(\emptyset)$ is the set of all nonnegative integers that has only digits 0 and 1 in its base 3 representation.

There are many natural ways to generalize this sequence. We cover two directions. The first is to vary a single equation that we wish to avoid, while the second is to vary the length of arithmetic progressions we wish to avoid. These are detailed below.

1.1. Greedy Sum-free Sequence. Avoiding a 3-AP amounts to avoiding solutions to $x + y = 2z$. One natural generalization is to study how varying the equation we wish to avoid to $x + y = z$ changes the problem.

Definition 3. Given a subset $A_0 = \{a_1 < a_2 < \dots < a_t\} \subseteq \mathbb{N}$ such that the equation $x + y = z$ does not have a solution in A_0 , the **greedy sum-free sequence** $T(A_0)$ is the sequence $\{a_1, a_2, \dots\}$ defined recursively by for all $i \geq t$, a_{i+1} is the smallest nonnegative integer $a > a_i$ such that $x + y = z$ does not have a solution in $\{a_1, \dots, a_i, a\}$.

Example 4. We have

$$T(\{3, 4\}) = \{3, 4, 5, 11, 12, 13, 19, 20, 21, 27, 28, 29, 35, 36, 37, \dots\},$$

which we note that it is the set of positive integers congruent to 3, 4, or 5 modulo 8.

It is conjectured by Erdős that the greedy sequence avoiding the solution to $x + y = z$ has a modulo n structure similar to the above, apart from some finite set of exceptions.

Conjecture 5 ([EG80; Blo26b]). For any $A_0 \subseteq \mathbb{N}$, if $T(A_0) = \{a_1 < a_2 < \dots < a_\ell\}$, then the sequence $d_k = a_k - a_{k-1}$ is eventually periodic.

In other words, Erdős asks if there exists a positive integer n and $S \subseteq \mathbb{Z}/n\mathbb{Z}$ such that $T(A_0)$ differs to the set $\{x \in \mathbb{N}_0 : x \bmod n \in S\}$ by a finite set. This question is open in the general case, but when the behavior of the initial terms is sufficiently easy to control, one can explicitly write down what occurs. Indeed, we prove that **Conjecture 5** holds for all sets A_0 of size 2 (and therefore of size 1, 0 as well).

Theorem 6. For any distinct positive integers $a < b$ with $b \neq 2a$, if $T(\{a, b\}) = \{a_1 < a_2 < \dots\}$, then the sequence $a_k - a_{k-1}$ is eventually periodic.

We also identify the period in each case. See **Section 2** for more details.

1.2. Growth Rate of Greedy k -AP Free Sequence. We consider the following generalization of Stanley sequences where we change constraints from being 3-AP free to being k -AP free.

Definition 7. Given a k -AP free subset $A_0 = \{a_1 < a_2 < \dots < a_t\} \subseteq \mathbb{N}_0$, the **generalized Stanley sequence** $S_k(A_0)$ is a set $\{a_1 < a_2 < \dots\}$ such that for all $i \geq t + 1$, a_i is the smallest nonnegative integer a such that $a > a_{i-1}$ and $\{a_1, \dots, a_{i-1}, a\}$ is k -AP free.

Example 8. The original Stanley sequence is the sequence

$$S_3(\{0\}) = \{0, 1, 3, 4, 9, 10, 12, 13, 27, 28, 30, 31, \dots\}.$$

On the other hand, if we change initial terms, we will get a different-looking sequence. For example,

$$S_3(\{0, 4\}) = \{0, 4, 5, 7, 8, 11, 12, 16, 23, 26, 31, 33, \dots\},$$

whose closed form is more obscure. Meanwhile changing the value of k will get a denser sequence. For example,

$$S_4(\{0\}) = \{0, 1, 2, 4, 5, 7, 8, 9, 14, 15, 16, 18, \dots\}.$$

We are interested in investigating the growth rate of this sequence as A_0 and k vary. The first result of this sort is a lower bound on density of $S_3(A_0)$, first proved by Gerver and Ramsey [GR79] and rediscovered by Moy [Moy11, Thm. 1.1].

Proposition 9 ([GR79; Moy11]). For any $A_0 \subseteq \mathbb{N}_0$ and $\varepsilon > 0$, if $A = S_3(A_0)$, then

$$|A \cap \{0, 1, \dots, n\}| > (\sqrt{2} - \varepsilon)\sqrt{n}$$

for all sufficiently large integers n .

In other words, the n -th element of $S_3(A_0)$ is $O(n^2)$. It is widely believed that the exponent 2 cannot be improved. In particular, the Erdős problem #271 [Blo26a] conjectures the following.

Conjecture 10 ([EG80; Blo26a]). For any positive integer n_0 not of the form 3^k or $2 \cdot 3^k$ for $k \in \mathbb{N}_0$, the n -th term of $S_3(\{0, n_0\})$ is $\Theta(n^2 / \log n)$.

Note that if n_0 is of the form 3^k or $2 \cdot 3^k$, then $S_3(\{0, n_0\})$ has a base-3 closed form similar to the closed form $S_3(\{0\})$, and the n -th term of $S_3(\{0, n_0\})$ is instead $\Theta(n^{\log_2 3})$. See [OS78] for more details.

We do not discuss or make progress on [Conjecture 10](#). However, we are interested in the growth rate of $S_k(A_0)$ for general $k \geq 4$. As in the case for $k = 3$, some initial sets lead to a sequence with a nice closed form. For example, [MR16, Lem. 6.4] proved that for any prime p , the set $S_p(\{0\})$ is the set of nonnegative integers whose base p representation does not contain digit $p - 1$. However, in general, we expect the sparsest possible sequence to look like random random and thus make the following conjecture.

Conjecture 11. For any $\varepsilon > 0$ and subset $A_0 \subseteq \mathbb{N}_0$, the n -th term of $S_k(A_0)$ is $O\left(n^{\frac{k-1}{k-2}+\varepsilon}\right)$. In other words, the conjecture states that for any $\varepsilon > 0$ and subset $A_0 \subseteq \mathbb{N}_0$

$$|S_k(A_0) \cap \{0, 1, \dots, n\}| > \Omega\left(n^{\frac{k-2}{k-1}-\varepsilon}\right)$$

In order to make progress on [Conjecture 11](#), we note that for any $x > \max A_0$, there exists $d \in \mathbb{N}_0$ such that $x - d, x - 2d, \dots, x - (k - 1)d$ are all in $S_k(A_0)$. This is useful for obtaining a lower bound on density of $S_k(A_0)$ because it forces many elements to be in $S_k(A_0)$. This motivates us to make the following definition.

Definition 12. A subset $A \subseteq \mathbb{N}_0$ is **k -AP covering** if there exists n_0 such that for all $x > n_0$, there exists $d \in \mathbb{N}_0$ such that $x - d, x - 2d, \dots, x - (k - 1)d$ are all in A .

The study of k -AP covering sets was initiated by Kiss, Sándor, and Yang [KSY18]. Their first observation is that the proof of [Proposition 9](#) directly generalizes to any 3-AP covering set. In particular, directly from the proof of [Proposition 9](#), we get the following.

Proposition 13. For any 3-AP covering set $A \subseteq \mathbb{N}_0$ and $\varepsilon > 0$, we have

$$|A \cap \{0, 1, \dots, n\}| > (\sqrt{2} - \varepsilon)\sqrt{n}$$

for all sufficiently large integer n .

Kiss, Sándor, and Yang also proved that this bound is tight up to constant factor, and their construction is subsequently improved by Chen and Fang [Che18; Fan21]. They also make the following conjecture for any k .

Conjecture 14 ([KSY18, Conj. 1]). There exists a constant $c_k > 0$ such that for any k -AP covering set $A \subseteq \mathbb{N}_0$, we have

$$|A \cap \{0, 1, \dots, n\}| > c_k n^{\frac{k-2}{k-1}}$$

for infinitely many nonnegative integers n .

Note that the exponent $\frac{k-2}{k-1}$ cannot be replaced by any larger exponent, as shown in [KSY18, Thm. 3] by picking a random set A , including each positive integer randomly and independently with a certain probability. [Conjecture 14](#) is saying that this random construction

is basically optimal. If [Conjecture 14](#) is true, we will get that the n -th term of $S_k(A_0)$ is $O\left(n^{\frac{k-2}{k-1}}\right)$, which translates to a bound on the growth rate of Stanley sequences.

Unfortunately, our result (proven in [Section 3.1](#)) shows that [Conjecture 14](#) is false for all $k \geq 6$, leaving only the case $k \in \{4, 5\}$ open.

Theorem 15. *For all positive integer $k \geq 6$, there exists a constant $\varepsilon = \varepsilon_k > 0$, and a k -AP covering set A such that*

$$|A \cap \{0, 1, \dots, n\}| < n^{\frac{k-2}{k-1} - \varepsilon}$$

for all sufficiently large positive integer n .

Even worse, we give a different construction in [Section 3.2](#) which gets a much better exponent for all sufficiently large k . This construction is a modification of a construction to a similar problem due to Green and Ruzsa [[GR19](#), Thm. 1.2]

Theorem 16. *There exists a constant $c > 0$ such that for all sufficiently large k , there exists a k -AP covering set A such that*

$$|A \cap \{0, 1, \dots, n\}| < n^{1 - \frac{c}{\log \log k}}$$

for all sufficiently large positive integers n .

1.3. Notations. We let $\mathbb{N}$ be the set of positive integers, $\mathbb{N}_0$ be the set of nonnegative integers, and $\mathbb{Z}/N\mathbb{Z}$ denote the ring of integers modulo N . A **k -AP** is an arithmetic progression of length k , which could be constant (i.e., all terms equal). A subset $A \subseteq \mathbb{N}_0$ is **k -AP free** if it does not contain a nonconstant k -AP.

In this paper (particularly in [Section 2](#)), we write $[j, k]$ to denote $\{x \in \mathbb{Z} \mid j \leq x \leq k\}$, and likewise with open and half-open intervals.

For any function $f : \mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{R}$, we let $\mathbb{E}_x[f(x)]$ denote the expected value of $f(x)$, where x is taken to be uniformly at random across elements of $\mathbb{Z}/N\mathbb{Z}$.

We use the standard asymptotics notation throughout this paper. In particular, we say that

- $f(n) = O(g(n))$ if there exists a constant $C > 0$ such that $f(n) \leq Cg(n)$ for all sufficiently large n .
- $f(n) = \Omega(g(n))$ if there exists a constant $C > 0$ such that $f(n) \geq Cg(n)$ for all sufficiently large n .
- $f(n) = \Theta(g(n))$ if there exist constants $C_1, C_2 > 0$ such that $C_1g(n) \leq f(n) \leq C_2g(n)$ for all sufficiently large n ; equivalently, $f(n) = \Theta(g(n))$ if and only if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$.

1.4. Organization of the Paper. There are three key results in the paper: [Theorem 6](#), [Theorem 15](#), and [Theorem 16](#). In [Section 2](#), we explore the periodicity of greedy sum-free sequence and prove [Theorem 6](#). In [Section 3](#), we construct a sparse k -AP covering set using two different methods. The first method, which is explored in [Section 3.1](#), uses Fourier analysis on $\mathbb{Z}/N\mathbb{Z}$, where N is a product of several prime numbers, to prove [Theorem 15](#). The second method, which is explored in [Section 3.2](#), uses number-theoretic construction to prove [Theorem 16](#).

2. PERIODICITY OF GREEDY SUM-FREE SEQUENCE

In this section, we prove [Theorem 6](#). Let $a < b$ be positive integers such that $2a \neq b$. We split into cases based on a and b .

In most cases, we will get many blocks of length a (i.e. of the form $[t, t + a)$). To this end, we will want to study which subsequent integers this prohibits from being part of the sequence.

Lemma 17. *Let $S = [x, y]$ for $x, y \in \mathbb{Z}$. Then each $z \in [2x, 2y]$ can be expressed as a sum of two elements in S and can be done so without using any single element t of S unless $z \in \{2x, 2x + 1\}$, when $t = x, x + 1$ or $z \in \{2y - 1, 2y\}$ when $t = y - 1, y$.*

Proof. Any $z \in [2x, 2y]$ is of the form $\lfloor z/2 \rfloor + \lceil z/2 \rceil = (\lfloor z/2 \rfloor - 1) + (\lceil z/2 \rceil + 1)$. We have $\lfloor z/2 \rfloor, \lceil z/2 \rceil \in [x, y]$ and $\lfloor z/2 \rfloor - 1 \in [x, y]$ when $z > 2x + 1$ and $\lceil z/2 \rceil + 1 \in [x, y]$ when $z < 2y - 1$. $\square$

With this result in hand, we first deal with a special case that is illustrative of the general approach.

Proof when $b = 2a - 1$. Suppose that $a \geq 4$. We claim the sequence is

a	$2a - 1$	$2a + 1, 2a + 2, \dots, 3a - 2$	$3a$	$4a - 1$
$6a - 1$	$7a - 2$	$7a, 7a + 1, \dots, 8a - 3$	$8a - 1$	$9a - 2$
$11a - 2$	$12a - 3$	$12a - 1, 12a, \dots, 13a - 4$	$13a - 2$	$14a - 3$

and continues with period $p = 5a - 1$. To see this, proceed by induction, and observe that

- (1) $kp + 2a = (kp + a) + a$
 - (2) $kp + (3a - 1) = (kp + a) + (2a - 1)$
 - (3) $kp + (3a + t) = (kp + a) + (2a + t)$ for $1 \leq t < a - 1$
 - (4) $kp + (4a + t)$ for $0 \leq t < 2a - 1$ can be written as an element in the middle three columns, plus kp plus another element in the middle three columns by Lemma 17.
- Note that here we use $a \geq 4$.

while $A = \{a, 2a - 1, [2a + 1, 3a - 2], 3a, 4a - 1\}$ is a subset of $\mathbb{Z}/p\mathbb{Z}$ with $A + A \cap A = \emptyset$.

When $a = 3, b = 5$, we can compute explicitly that $a_n = 2n + 1$ (which does repeat with period 14) and when $a = 2, b = 3$ we fall into the $a + 1 = b$ case and have (minimal) period 5, which is not a divisor of 9. $\square$

For the remaining cases, we will identify the structure and periodicity of the sequence. Proving that these amount to the exact structure of the sequence a_n is left as an induction exercise that is a slight modification of the above proof.

Proof when $a + 1 \leq b \leq \frac{3}{2}a$. Note that $a > 2$. The sequence starts

$$a, b, b + 1, \dots, \widehat{2a}, \dots, a + b - 1.$$

Now, we claim the next term is $2(a + b - 1) + 1 = 2a + 2b - 1$. To see this, observe that $z \in [a + b, 2a + b - 1]$ cannot be the next term, except possibly $3a$ as they are of the form $a + [b, a + b - 1]$. Now observe $3a = (3a - b) + b$, and

$$3a - b = 2a \iff b = a \text{ and } 3a - b \geq b \iff 3a \geq 2b \text{ and } 3a - b < a + b \iff 2a < 2b.$$

Now $b + b = 2b \leq 2a + b - 1$ and by Lemma 17 with $x = b, y = a + b - 1$, the next term cannot be $\leq 2a + 2b - 2$ unless $2a = b, 2a = b + 1, 2a = a + b - 1$ or $2a = a + b - 2$. Of course, the first two cannot happen as $a > 2$ and $b \leq \frac{3}{2}a$.

• Suppose $b > a + 2$. Then the next valid elements are $[2a + 2b - 1, 3a + 2b - 1)$, so we begin

$$a, \quad b, b + 1, \dots, \widehat{2a}, \dots, a + b - 1,$$

$$b + (2a + b - 1), b + (2a + b - 1) + 1, \dots, b + (2a + b - 1) + (a - 1), \\ b + 2(2a + b - 1), b + 2(2a + b - 1) + 1, \dots, b + 2(2a + b - 1) + (a - 1),$$

and consequently repeat with period $2a + b - 1$.

- In the case $b = a + 1$, the sequence contains the elements $[a, 2a)$ and then $[4a - 1, 5a - 1)$, repeating with period $3a - 1 = 2a + b - 2$

- Finally, in the case $b = a + 2$, the sequence begin

$$a, a + 2, a + 3, \dots, 2a - 1, 2a + 1, 4a + 1, 4a + 3, 4a + 4, \dots, \widehat{5a + 1}, \dots,$$

and thus repeats with period $3a + 1 = 2a + b - 1$. $\square$

Proof when $\frac{3}{2}a < b < 2a - 1$. Like in the prior case, the sequence starts

$$a, b, b + 1, \dots, \widehat{2a}, \dots, a + b - 1, 3a.$$

For $z \in b + [b, a - b)$, the only term that could be included is $b + 2a$ which can also be written as $(b + 1) + (2a - 1)$ and since $b + 1 < 2a$, this prohibits $b + 2a$.

The next possible subsequent element is $2(a + b - 1) + 1 = 2b + 2a - 1 = (3a) + (2b - 1 - a)$. Now note that $2b - 1 - a \geq b$ and $2b - 1 - a \leq a + b - 1 \iff b \leq 2a$, so $2b - 1 - a \in A$ and thus $2b + 2a - 1 \notin A$.

Next, $z \in 3a + [b, a - b - 1)$ are not valid, so the next valid element is $5a$ exactly when

$$5a > 2(a + b - 1) \iff 3a + 2 > 2b.$$

If $3a + 2 = 2b + 1$ then $5a$ is valid, so the sequence starts

$$a, b, b + 1, \dots, \widehat{2a}, \dots, a + b - 1, 3a, \\ 5a, 4a + b, 4a + b + 1, \dots, \widehat{6a}, \dots, 5a + b - 1, \dots,$$

and continues with period $4a$.

Otherwise, the next valid term is $4a + b$, so the sequence starts

$$a, \quad b, b + 1, \dots, \widehat{2a}, \dots, a + b - 1, 3a, \\ 4a + b, 4a + b + 1, \dots, \widehat{6a}, \dots, 5a + b - 1, 7a,$$

and continues with period $4a$, and thus either way the sequence is periodic with period $4a$. $\square$

Proof when $b > 2a$. Suppose $t \geq 2$ is the unique integer with $ta < b \leq (t + 1)a$. Suppose $a > 2$. The sequence starts

$$a, b, b + 1, \dots, b + (a - 1), b + 2a, b + 2a + 1, \dots, b + 2a + (a - 1).$$

It will continue with blocks of length a , of the form $[b + 2na, b + (2n + 1)a)$ until $2b$. We now split into cases based on the parity of t .

- Suppose $t = 2k$ is even. The sequence starts

$$a, b, b + 1, \dots, b + (a - 1), \\ b + 2a, \dots, b + 2a + (a - 1), \\ b + 4a, \dots, b + 4a + (a - 1), \\ \vdots \\ b + (2k)a, b + (2k)a + 1, \dots, 2b - 1.$$

From here, every element $z \in [2b, 4b - 2]$ except those congruent to $2b - 1$ modulo $2a$ can be written as a sum of elements of the form $b + s$ plus $z - (b + s)$, where

$s \in [0, a) \cap \mathbb{Z}$ and therefore $z - (b + s)$ is congruent to something less than a modulo $2a$ for some choice of s . Thus, the sequence continues $2b - 1 + 2na$ for $n = 1, \dots, t - 1$.

Then, $[2b - 1] + [b + (2k - 2)a + (a - 1) + 1] = 3b + (t - 1)a - 1$ is the next acceptable term, wherein we then repeat with period $2b + (t - 1)a - 1$.

- Alternatively, suppose $t = 2k + 1$ is odd. Then the sequence starts

$$\begin{aligned} & a, b, b + 1, \dots, b + (a - 1), \\ & \quad b + 2a, \dots, b + 2a + (a - 1), \\ & \quad b + 4a, \dots, \\ & \quad \vdots \\ & \quad b + (2k)a, b + (2k)a + 1, \dots, b + (2k)a + (a - 1). \end{aligned}$$

From the same logic as above, we get terms $2b - 1 + 2na$ for $n = 1, \dots, t - 2$. The next acceptable term is then $[b + (2k - 2)a + (a - 1)] + [2b - 1 + 2a] + 1 = 3b + (2k + 1)a - 1 = 3b + ta - 1$, wherein we then repeat with period $2b + ta - 1$. $\square$

The remaining cases when $a = 1, a = 2$ can be computed explicitly. This results in the following table.

Period	Conditions
$3a - 1 = 2a + b - 2$	$a + 1 = b$
$2a + b - 1$	$a + 1 < b \leq \frac{3}{2}a$
$4a$	$\frac{3}{2}a < b < 2a - 1$
$5a - 1$	$b = 2a - 1, a > 2$
$3b - 1$	$a = 2, b = 4k + 2$
$4b - 1$	$a = 1, b = 2k > 2$
$(2k - 1)a + 2b - 1$	$(2k - 1)a < b \leq (2k + 1)a$, except above

TABLE 1. Periods and conditions upon which they occur

3. CONSTRUCTING SPARSE k -AP COVERING SETS

In this section, we prove [Theorem 15](#) and [Theorem 16](#) by giving two different ways to construct a sparse k -AP covering set. The first way, used to prove [Theorem 15](#), is based on Fourier-analytic construction in $\mathbb{Z}/N\mathbb{Z}$, where N is suitably chosen. The second way constructs A as a union of sets in form $\{x - d(x), x - 2d(x), \dots, x - (k - 1)d(x)\}$, where $d(x)$ is a function of x , chosen suitably so that the union is confined within a smaller range of positive integers. This second construction is a close modification of the proof of [\[GR19, Thm. 1.2\]](#).

3.1. Fourier-Analytic Construction: Proof of [Theorem 15](#). Throughout this section, we fix an integer $k \geq 6$. Before delving into the detail, let us briefly outline the whole proof. The proof of [Theorem 15](#) proceeds in 5 steps.

- (1) First, we construct a function $f_p : \mathbb{Z}/p\mathbb{Z} \rightarrow (0, \infty)$ such that $\mathbb{E}_x[f_p(x)] = 1$ and $\mathbb{E}_x[f_p(a - 3x)f_p(a - 4x)f_p(a - 5x)] > 1.001$ for all $a \in \mathbb{Z}/p\mathbb{Z}$ using a Fourier-analytic construction.
- (2) Using the Chinese remainder theorem, we construct a function $f : \mathbb{Z}/Q\mathbb{Z} \rightarrow (0, \infty)$ (where Q is a product of several primes) such that $\mathbb{E}_x[f(x)] = 1$ but

$$\mathbb{E}_x[f(a - 3x)f(a - 4x)f(a - 5x)] > 100^{k^2} \log Q,$$

giving a $\log Q$ -factor and a constant factor gain. The $\log Q$ factor is necessary for the step (4) to work.

- (3) By modifying f , we construct a function $g : \mathbb{Z}/Q\mathbb{Z} \rightarrow (0, \infty)$ such that $\mathbb{E}_x[g(x)] = 1$ and

$$\mathbb{E}_x \left[\prod_{j=1}^{k-1} g(a - jx) \right] > 50^{k^2} \log Q.$$

It is worth noting that we are sacrificing some constant factor gain from step (2) in order to control other elements $a - jx$ for $j \notin \{3, 4, 5\}$. This does not affect our construction since the gain in step (2) can be made arbitrarily large.

- (4) By carefully sampling according to distribution determined by g , we construct a subset $B \subseteq \{0, 1, \dots, N-1\}$ such that $|B| < \frac{1}{2}N^{\frac{k-2}{k-1}}$ and for any $a \in \{0, 1, \dots, N-1\}$, there exists x such that $a - x, a - 2x, \dots, a - (k-1)x$ are all in B .
- (5) Finally, by the tensor power trick, we construct the desired k -AP covering set A .

Below, each step will be presented as a lemma, and the formal statements can be found there. The following lemma handles step (1).

Lemma 18. *For all prime $p > 100$, there exists a function $f_p : \mathbb{Z}/p\mathbb{Z} \rightarrow \mathbb{R}$ such that*

- (a) $0 < f_p(x) < 2$ for all $x \in \mathbb{Z}/p\mathbb{Z}$.
- (b) $\mathbb{E}_x[f_p(x)] = 1$.
- (c) For any $a \in \mathbb{Z}/p\mathbb{Z}$, we have

$$\mathbb{E}_x[f_p(a - 3x)f_p(a - 4x)f_p(a - 5x)] > 1.001.$$

Proof. Let $\omega = \exp(2\pi i/p)$. We take function f_p of the form

$$f_p(x) = 1 + 0.2(\omega^x + \omega^{-x}) + 0.1(\omega^{2x} + \omega^{-2x}),$$

so that $f_p(x)$ is indeed a real number in $(0, 2)$, satisfying (a). Since $\mathbb{E}_x[\omega^{bx}] = 0$ for all b not divisible by p , we get by that $\mathbb{E}_x[f_p(x)] = 1$, verifying (b).

Now, we have to verify (c). To do this, we note that after full expansion of $f_p(a - 3x)f_p(a - 4x)f_p(a - 5x)$, we get terms that are linear combination of $\omega^{\varepsilon_1(a-3x) + \varepsilon_2(a-4x) + \varepsilon_3(a-5x)}$ for some $\varepsilon_1, \varepsilon_2, \varepsilon_3 \in \{0, 1, -1, 2, -2\}$. If $3\varepsilon_1 + 4\varepsilon_2 + 5\varepsilon_3 \neq 0$, then we get a multiple of $\mathbb{E}_x[\omega^{bx}]$ for some integer b with $|b| < 100$, so such terms vanish. By checking all 5^3 combinations of $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$, the only combinations that make $3\varepsilon_1 + 4\varepsilon_2 + 5\varepsilon_3 = 0$ are

$$(\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{(0, 0, 0), (1, -2, 1), (-1, 2, -1), (2, 1, -2), (-2, 1, -2)\}.$$

The combinations $(1, -2, 1)$ and $(-1, 2, -1)$ each contribute $(0.2)^2(0.1)$ to the sum. The combinations $(2, 1, -2)$ and $(-2, 1, -2)$ contributes $(0.2)(0.1)^2\omega^a$ and $(0.2)(0.1)^2\omega^{-a}$ to the sum, respectively. Therefore, we get that

$$\mathbb{E}_x[f_p(a - 3x)f_p(a - 4x)f_p(a - 5x)] = 1.008 + 0.002(\omega^a + \omega^{-a}) > 1.001,$$

as desired. $\square$

Next, we handle step (2) by using Chinese remainder theorem to stitch together f_p for several primes p .

Lemma 19. *There are infinitely many positive integers Q and a function $f : \mathbb{Z}/Q\mathbb{Z} \rightarrow (0, \infty)$ such that*

- (a) $0 < f(x) < Q^{\frac{1}{100k}}$ for all $x \in \mathbb{Z}/Q\mathbb{Z}$.
- (b) $\mathbb{E}_x[f(x)] = 1$

(c) for any $a \in \mathbb{Z}/Q\mathbb{Z}$, we have

$$\mathbb{E}_x[f(a-3x)f(a-4x)f(a-5x)] > 100^{k^2} \log Q.$$

Proof. Let $p_1, p_2, \dots, p_n$ be the first n primes greater than 2^{100k} , where n is sufficiently large (to be specified later). Take $Q = p_1 \cdots p_n$ and

$$f(x) = f_{p_1}(x) \cdots f_{p_n}(x),$$

where f_{p_i} are as in Lemma 18. Since $f_{p_i}(x) < 2 < p_i^{\frac{1}{100k}}$, we get that $f(x) < Q^{\frac{1}{100k}}$, verifying (a).

To verify (b), we note by Chinese remainder theorem that if x is uniform in $\mathbb{Z}/Q\mathbb{Z}$, then the variables $x \bmod p_1, \dots, x \bmod p_n$ can be treated as independent variables. This implies that

$$\mathbb{E}_x[f(x)] = \mathbb{E}_x[f_{p_1}(x)] \cdots \mathbb{E}_x[f_{p_n}(x)] = 1,$$

verifying (b).

Finally, to verify (c), we note by the same Chinese remainder theorem argument that

$$\begin{aligned} \mathbb{E}_x[f(a-3x)f(a-4x)f(a-5x)] &= \prod_{i=1}^n \mathbb{E}_x[f_{p_i}(a-3x)f_{p_i}(a-4x)f_{p_i}(a-5x)] \\ &> 1.001^n. \end{aligned}$$

By a variant of the prime number theorem, the product of first n primes greater than 2^{100k} is $\exp(O(n \log n))$. Thus, we get that $\log Q = O(n \log n)$, which is less than $100^{-k^2} 1.001^n$ for all n . Thus, $1.001^n > 100^{k^2} \log Q$ if n is sufficiently large, implying existence of such Q . $\square$

We now handle step (3) by smoothing out function f , yielding a function g that controls $a - jx$ for all $j \in \{1, 2, \dots, k-1\}$ (at the cost of losing a constant factor from the bound, but this loss is mitigated from the amplification in step (2)). This is where we use $k \geq 6$.

Lemma 20. *For each Q in Lemma 19, there exists a function $g : \mathbb{Z}/Q\mathbb{Z} \rightarrow (0, \infty)$ such that*

- (a) $0 < g(x) < Q^{\frac{1}{100k}}$ for all $x \in \mathbb{Z}/Q\mathbb{Z}$.
- (b) $\mathbb{E}_x[g(x)] = 1$
- (c) for any $a \in \mathbb{Z}/Q\mathbb{Z}$, we have

$$\mathbb{E}_x \left[\prod_{j=1}^{k-1} g(a-jx) \right] > 50^{k^2} \log Q.$$

Proof. We take $g(x) = \frac{1+f(x)}{2}$, where f as in Lemma 19, which clearly satisfies (a) and (b). To verify (c), we use the bounds $g(a-jx) \geq \frac{f(x)}{2}$ for $j \in \{3, 4, 5\}$ and $g(a-jx) \geq \frac{1}{2}$ for $j \notin \{3, 4, 5\}$, giving

$$\begin{aligned} \mathbb{E}_x \left[\prod_{j=1}^{k-1} g(a-jx) \right] &\geq \mathbb{E}_x \left[\frac{1}{2^{k-1}} f(a-3x)f(a-4x)f(a-5x) \right] \\ &\geq \frac{1}{2^{k-1}} 100^{k^2} \log Q > 50^{k^2} \log Q. \end{aligned} \quad \square$$

We now sample from g to get a subset of integers.

Lemma 21. *There exist a positive integer N and a set $B \subseteq [N]$ such that*

- (a) $|B| < \frac{1}{2} N^{\frac{k-2}{k-1}}$ and

(b) for any $x \in \{0, 1, \dots, N\}$, there exists a nonnegative integer d such that $x - d, x - 2d, \dots, x - (k-1)d$ are all in B .

Proof. We take Q and g as in Lemma 20. Let $N = 10k^2Q^2$. We construct the set B as follows:

- include each $x \in [1, 10k^2Q]$;
- include each $x \in [10k^2Q + 1, 20k^2Q]$ with probability $\frac{1}{100k^2}Q^{-\frac{1}{k-1}}g(x)$;
- include each $x \in [20k^2Q + 1, 30k^2Q]$ with probability $\frac{1}{100k^2}(2Q)^{-\frac{1}{k-1}}g(x)$;
- include each $x \in [30k^2Q + 1, 40k^2Q]$ with probability $\frac{1}{100k^2}(3Q)^{-\frac{1}{k-1}}g(x)$;
- $\vdots$
- include each $x \in [10(Q-1)k^2Q + 1, 10k^2Q^2]$ with probability $\frac{1}{100k^2}((Q-1)Q)^{-\frac{1}{k-1}}g(x)$.

Note that

$$\begin{aligned}
\mathbb{E}[|B|] &= 10k^2Q + \left(1 + 2^{-\frac{1}{k-1}} + 3^{-\frac{1}{k-1}} + \dots + (Q-1)^{-\frac{1}{k-1}}\right) \cdot \left(\frac{1}{100k^2}Q^{-\frac{1}{k-1}}\right) \sum_{x=0}^{10k^2Q-1} f(x) \\
&< 20k^2Q + \left(\frac{k-1}{k-2} \cdot Q^{\frac{k-2}{k-1}}\right) \cdot \left(\frac{1}{100k^2}Q^{-\frac{1}{k-1}}\right) \cdot (10k^2Q \cdot \mathbb{E}_x[g(x)]) \\
&< 20k^2Q + \frac{1}{5}Q^{2 \cdot \frac{k-2}{k-1}} \\
&< \frac{1}{4}Q^{2 \cdot \frac{k-2}{k-1}},
\end{aligned}$$

if Q is sufficiently large. Thus, by Markov's inequality, we deduce that

$$\mathbb{P}\left(|B| > \frac{1}{2}N^{\frac{k-2}{k-1}}\right) < 0.1.$$

Next, we look at the second condition. For any $a \in [10k^2nQ + 1, 10k^2(n+1)Q]$, we can pick any x from the interval $[knQ, (k+1)nQ]$. We claim that for any distinct $x_1, x_2 \in [knQ, (k+1)nQ]$, the sets $\{a-x_1, a-2x_1, \dots, a-(k-1)x_1\}$ and $\{a-x_2, a-2x_2, \dots, a-(k-1)x_2\}$ are disjoint. Assume that they are not, then there exists $u, v \in \{1, 2, \dots, k-1\}$ such that $ux_1 = vx_2$, so $\frac{x_1}{x_2} = \frac{u}{v}$. However, $\frac{x_1}{x_2}$ is between $\frac{k}{k+1}$ and $\frac{k+1}{k}$, which cannot be equal to $\frac{u}{v}$ for $u, v \in \{1, 2, \dots, k-1\}$ (unless $u = v$).

Thus, we deduce that the events of the form $\{a-x, a-2x, \dots, a-(k-1)x\} \subseteq A$ are independent across all x . Therefore, the probability that a does not satisfies the second condition is

$$\begin{aligned}
\mathbb{P}(a \text{ fail}) &= \prod_{x=knQ}^{(k+1)nQ} \left(1 - \prod_{j=1}^{k-1} \mathbb{P}[a-jx \in B]\right) \\
&= \prod_{x=1}^{nQ} \left(1 - \prod_{j=1}^{k-1} \left(\frac{1}{100k^2} \cdot (nQ)^{-\frac{1}{k-1}} g(a-jx)\right)\right) \\
&= \prod_{x=1}^{nQ} \left(1 - \frac{1}{(100k^{2k})nQ} \prod_{j=1}^{k-1} g(a-jx)\right) \\
&< \prod_{x=1}^{nQ} \left(1 - \frac{1}{10k^2nQ} \prod_{j=1}^{k-1} g(a-jx)\right)
\end{aligned}$$

$$\begin{aligned}
&\leq \prod_{x=1}^n \exp \left(-\frac{1}{10^{k^2} n Q} \prod_{j=1}^{k-1} g(a - jx) \right) && \text{(using } \exp(-x) \geq 1 - x \text{)} \\
&\leq \exp \left(-\frac{1}{10^{k^2} n Q} \sum_{x=1}^{nQ} \prod_{j=1}^{k-1} g(a - jx) \right) \\
&\leq \exp \left(-\frac{1}{10^{k^2}} \mathbb{E}_x \left[\prod_{j=1}^{k-1} g(a - jx) \right] \right) \\
&\leq \exp \left(-100 \log Q \right) < \frac{1}{Q^{100}},
\end{aligned}$$

so summing across all $a \in [N]$ and using the union bound gives that the probability that the second condition fails is at most $\frac{1}{2}$. Thus, by the union bound, there is a nonzero probability that B satisfies both condition. Therefore, such B exist. $\square$

Finally, we now prove [Theorem 15](#), which we reproduce below.

Theorem 15. *For all positive integer $k \geq 6$, there exists a constant $\varepsilon = \varepsilon_k > 0$, and a k -AP covering set A such that*

$$|A \cap \{0, 1, \dots, n\}| < n^{\frac{k-2}{k-1}-\varepsilon}$$

for all sufficiently large positive integer n .

Proof of Theorem 15. Take N and B as in [Lemma 21](#). We let

$$A = \left\{ \sum_{i=0}^{\infty} e_i (N+1)^i : e_i \in B \right\}$$

be the set of positive integers whose base $(N+1)$ representations have all digits in B . Clearly, A is k -AP covering since we can take an AP corresponding to each digit that covers each element in A . Note that

$$|A \cap \{0, 1, \dots, n\}| \geq |B|^{\log_N n - 1} > \frac{1}{|B|} n^{\frac{\log |B|}{\log N}},$$

and since $\frac{\log |B|}{\log N} < \frac{k-2}{k-1}$, it would suffice to pick any $\varepsilon > 0$ such that $\frac{\log |B|}{\log N} < \frac{k-2}{k-1} - \varepsilon$. $\square$

3.2. Arithmetic Construction: Proof of Theorem 16. In this subsection, we prove [Theorem 16](#). The proof of this theorem is very similar to Green and Rusza result [[GR19](#), Thm. 1.2]. In particular, we start by construct a subset that satisfies the condition for $\{0, 1, \dots, Q-1\}$, where Q is a product of several primes. This set is a union of sets of the form $\{x - d(x), x - 2d(x), \dots, x - (k-1)d(x)\}$, where $d(x) = x^2 \bmod Q$. This is so that each $x - jd(x)$ is confined in a small number of residue classes modulo Q , which make B small enough to meet the required bound.

Finally, we use the base- Q representation construction (similar to the last step of the proof of [Theorem 15](#)) to get the desired AP-covering set.

The following lemma constructs such subset of $\{0, 1, \dots, Q-1\}$.

Lemma 22. *There exists a constant $c > 0$ such that for all sufficiently large k , there exists a positive integer Q and a subset $B \subseteq \{0, 1, \dots, N-1\}$ such that*

$$|B| < N^{1 - \frac{c}{\log \log k}}$$

and for every $x \in \{0, 1, \dots, N-1\}$, there exists a nonnegative integer d such that $x-d, x-2d, \dots, x-(k-1)d$ are all in B .

Proof. Let $m = \lceil 10 \log k \rceil$, $3 = p_1 < p_2 < \dots$ be the odd primes, and $Q = p_1 p_2 \dots p_m$. We take $N = Q^2$ and

$$B = \{0, 1, \dots, kQ - 1\} \cup \bigcup_{x=kQ}^{Q^2-1} \{x - d(x), x - 2d(x), \dots, x - (k-1)d(x)\}$$

where $d(x) = x^2 \bmod Q$ (the remainder when x^2 is divided by Q). Clearly, this satisfies the second condition, so it suffices to bound the size of the set. To bound the size of the union, we note that for all j and prime p_i for $1 \leq i \leq m$, we have that

$$x - j d(x) \equiv x - j x^2 \equiv \frac{1}{4j} - j \left(x - \frac{1}{2j} \right)^2 \pmod{p_i},$$

so $x - j d(x)$ takes at most $\frac{p_i+1}{2}$ residues in modulo p_i . Therefore, in modulo Q , it takes at most

$$\prod_{\substack{i \in \{1, 2, \dots, m\} \\ p_i \nmid j}} \frac{p_i + 1}{2} \cdot \prod_{\substack{i \in \{1, 2, \dots, m\} \\ p_i \mid j}} p_i \leq \frac{Q}{2^{m-\omega(j)}} \prod_{i=1}^m \left(1 + \frac{1}{p_i} \right) \leq \frac{Q}{2^{m/2}} \cdot k \leq \frac{Q}{k^4}$$

residues modulo Q . Here we use the bounds $\omega(j) = O\left(\frac{\log k}{\log \log k}\right) \ll m$ and $\prod_{i=1}^m \left(1 + \frac{1}{p_i}\right) = O(\log m) \ll k$.

Thus, we conclude that $x - j d(x)$ takes at most $\frac{Q}{k^4}$ distinct residues modulo Q . Taking union across all $j \in \{1, 2, \dots, k-1\}$ gives that the union in the definition of B takes at most $\frac{Q}{k^3}$ distinct residues modulo Q . Since every value is at most Q^2 , we get that there are at most $\frac{Q^2}{k^3}$ possible values. Thus,

$$|B| \leq kQ + \frac{Q^2}{k^3}.$$

Next, we note that by a variant of prime number theorem, $Q = \exp(\Theta(m \log m))$. Thus, $Q = k^{\Theta(\log \log k)}$. In particular, $Q > \frac{1}{2} k^4$ for all sufficiently large k , so

$$|B| \leq \frac{2Q^2}{k^3} < Q^{2 - \frac{2c}{\log \log k}} < N^{1 - \frac{c}{\log \log k}}$$

for some constant $c > 0$, implying that B satisfies the first condition. $\square$

We now prove [Theorem 16](#), which we reproduce below.

Theorem 16. *There exists a constant $c > 0$ such that for all sufficiently large k , there exists a k -AP covering set A such that*

$$|A \cap \{0, 1, \dots, n\}| < n^{1 - \frac{c}{\log \log k}}$$

for all sufficiently large positive integers n .

Proof of Theorem 16. Take N and B as in [Lemma 22](#). We let

$$A = \left\{ \sum_{i=0}^{\infty} e_i N^i : e_i \in B \right\}$$

be the set of positive integers whose base N representations have all digits in B . Clearly, A is k -AP covering since we can take an AP corresponding to each digit that covers each element in A . Note that

$$|A \cap \{0, 1, \dots, n\}| \geq |B|^{\log_N n - 1} > \frac{1}{|B|} n^{1 - \frac{c}{\log \log k}},$$

which is greater than $n^{1 - \frac{c'}{\log \log k}}$ for any $c' < c$ and sufficiently large n . $\square$

ACKNOWLEDGMENT AND WORK DISTRIBUTION

This paper was written for MIT Project Lab in Mathematics (18.821). The authors thank their mentors Wei Zhang and Matthew Halm for helpful discussion and feedback for the paper.

The first author came up with the proof and wrote up [Section 2](#), while the second author came up with the proof and wrote up [Section 3](#). Both authors worked together on the introduction.

REFERENCES

- [Beh46] F. A. Behrend. “On sets of integers which contain no three terms in arithmetical progression”. In: *Proc. Nat. Acad. Sci. U.S.A.* 32 (1946), pp. 331–332. ISSN: 0027-8424. DOI: [10.1073/pnas.32.12.331](https://doi.org/10.1073/pnas.32.12.331). URL: <https://doi.org/10.1073/pnas.32.12.331> (↑ p. 1).
- [Blo26a] Thomas F. Bloom. *Erdős Problem #271*. Accessed 17 Apr 2026. 2026. URL: <https://www.erdosproblems.com/271> (↑ p. 3).
- [Blo26b] Thomas F. Bloom. *Erdős Problem #341*. Accessed 21 Apr 2026. 2026. URL: <https://www.erdosproblems.com/341> (↑ p. 2).
- [Che18] Yong-Gao Chen. “On AP_3 -covering sequences”. In: *C. R. Math. Acad. Sci. Paris* 356.2 (2018), pp. 121–124. ISSN: 1631-073X,1778-3569. DOI: [10.1016/j.crma.2017.12.013](https://doi.org/10.1016/j.crma.2017.12.013). URL: <https://doi.org/10.1016/j.crma.2017.12.013> (↑ p. 3).
- [EG80] Paul Erdős and Ronald L. Graham. *Old and New Problems and Results in Combinatorial Number Theory*. Vol. 28. Monographies de L’Enseignement Mathématique. Geneva: L’Enseignement Mathématique, 1980 (↑ pp. 2, 3).
- [Fan21] Jin-hui Fang. “A note on AP_3 -covering sequences”. In: *Period. Math. Hungar.* 83.1 (2021), pp. 67–70. ISSN: 0031-5303,1588-2829. DOI: [10.1007/s10998-020-00350-1](https://doi.org/10.1007/s10998-020-00350-1). URL: <https://doi.org/10.1007/s10998-020-00350-1> (↑ p. 3).
- [GR79] Joseph L. Gerver and L. Thomas Ramsey. “Sets of integers with no long arithmetic progressions generated by the greedy algorithm”. In: *Math. Comp.* 33.148 (1979), pp. 1353–1359. ISSN: 0025-5718,1088-6842. DOI: [10.2307/2006472](https://doi.org/10.2307/2006472). URL: <https://doi.org/10.2307/2006472> (↑ p. 2).
- [GR19] Ben Green and Imre Z. Ruzsa. “On the arithmetic Kakeya conjecture of Katz and Tao”. In: *Period. Math. Hungar.* 78.2 (2019), pp. 135–151. ISSN: 0031-5303,1588-2829. DOI: [10.1007/s10998-018-0270-z](https://doi.org/10.1007/s10998-018-0270-z). URL: <https://doi.org/10.1007/s10998-018-0270-z> (↑ pp. 4, 7, 11).
- [KM23] Zander Kelley and Raghu Meka. “Strong bounds for 3-progressions”. In: *2023 IEEE 64th Annual Symposium on Foundations of Computer Science—FOCS 2023*. IEEE Computer Soc., Los Alamitos, CA, [2023] ©2023, pp. 933–973. ISBN: 979-8-3503-1894-4. DOI: [10.1109/FOCS57990.2023.00059](https://doi.org/10.1109/FOCS57990.2023.00059). URL: <https://doi.org/10.1109/FOCS57990.2023.00059> (↑ p. 1).
- [KSY18] S. Z. Kiss, Cs. Sándor, and Q.-H. Yang. “On generalized Stanley sequences”. In: *Acta Math. Hungar.* 154.2 (2018), pp. 501–510. ISSN: 0236-5294,1588-2632. DOI: [10.1007/s10474-018-0791-1](https://doi.org/10.1007/s10474-018-0791-1). URL: <https://doi.org/10.1007/s10474-018-0791-1> (↑ p. 3).
- [Moy11] Richard A. Moy. “On the growth of the counting function of Stanley sequences”. In: *Discrete Math.* 311.7 (2011), pp. 560–562. ISSN: 0012-365X,1872-681X. DOI: [10.1016/j.disc.2010.12.019](https://doi.org/10.1016/j.disc.2010.12.019). URL: <https://doi.org/10.1016/j.disc.2010.12.019> (↑ p. 2).

- [MR16] Richard A. Moy and David Rolnick. “Novel structures in Stanley sequences”. In: *Discrete Math.* 339.2 (2016), pp. 689–698. ISSN: 0012-365X,1872-681X. DOI: [10.1016/j.disc.2015.10.017](https://doi.org/10.1016/j.disc.2015.10.017). URL: <https://doi.org/10.1016/j.disc.2015.10.017> (↑ p. 3).
- [OS78] Andrew M. Odlyzko and Richard P. Stanley. “Some curious sequences constructed with the greedy algorithm”. In: 1978. URL: <https://api.semanticscholar.org/CorpusID:3262003> (↑ p. 3).

DEPARTMENT OF MATHEMATICS, MASSACHUSETTS INSTITUTE OF TECHNOLOGY, CAMBRIDGE, MA 02139,
USA

Email address: `pico@mit.edu`

DEPARTMENT OF MATHEMATICS, MASSACHUSETTS INSTITUTE OF TECHNOLOGY, CAMBRIDGE, MA 02139,
USA

Email address: `psaeng@mit.edu`